LIVE PRODUCTION SYSTEM | RENDER WEB SERVICE | EXPLAINABLE AI (XAI)

DATE: Nov 2026
STATUS: ACTIVE
200 OK

EcoVision AI: Integrated Watershed & Soil Erosion Command Center

Multi-Day Predictive Meteorological Forecasting, Game-Theoretic TreeSHAP Attribution, and RUSLE Physical Erosion Risk Triage

PUBLIC CLOUD WEB SERVICE
https://nitika-ma-am.onrender.com/
• Deployed & Verified (Render Fast-Boot Container)

GITHUB CODE REPOSITORY
https://github.com/Pratyaksh82/Nitika_ma-am
• Branch: main • Commit: 540f28a • Synchronized

EXECUTIVE SYSTEM ABSTRACT

EcoVision AI is a dual-engine environmental intelligence platform engineered to bridge machine learning rainfall forecasting with deterministic soil conservation physics. Deployed across the central Betwa-Narmada watershed divide (Raisen, Bhopal, Begumganj, Gairatganj, Silvani, Bareli, Udaipur, and Berasia), the system computes real-time catchment-scale erosion hazards using the Revised Universal Soil Loss Equation (RUSLE) parameterized by high-temporal XGBoost and Random Forest predictive precipitation models. Crucially, the platform integrates TreeSHAP game-theoretic attribution to provide auditors, hydrologists, and agricultural extension officers with transparent mathematical receipts for every hazard forecast.

PILLAR 1
Catchment Radar
GeoJSON polygon boundaries with real-time telemetry stream and station pins.

PILLAR 2
7-Day Trajectory
Autoregressive precipitation forecasting with What-If sensitivity sliders.

PILLAR 3
TreeSHAP Receipts
Game-theoretic feature attributions, waterfall charts, and beeswarm distributions.

PILLAR 4
RUSLE Interventions
Soil loss equation ($A=R \cdot K \cdot LS \cdot C \cdot P$) with dial gauge and agronomist bulletin.

END-TO-END ARCHITECTURAL PIPELINE

1. TELEMETRY INGEST
IMD Weather Feeds + Catchment Coordinates

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2. ML ENSEMBLE
XGBoost / RF (R^2: 0.908, MAE: 3.12mm)

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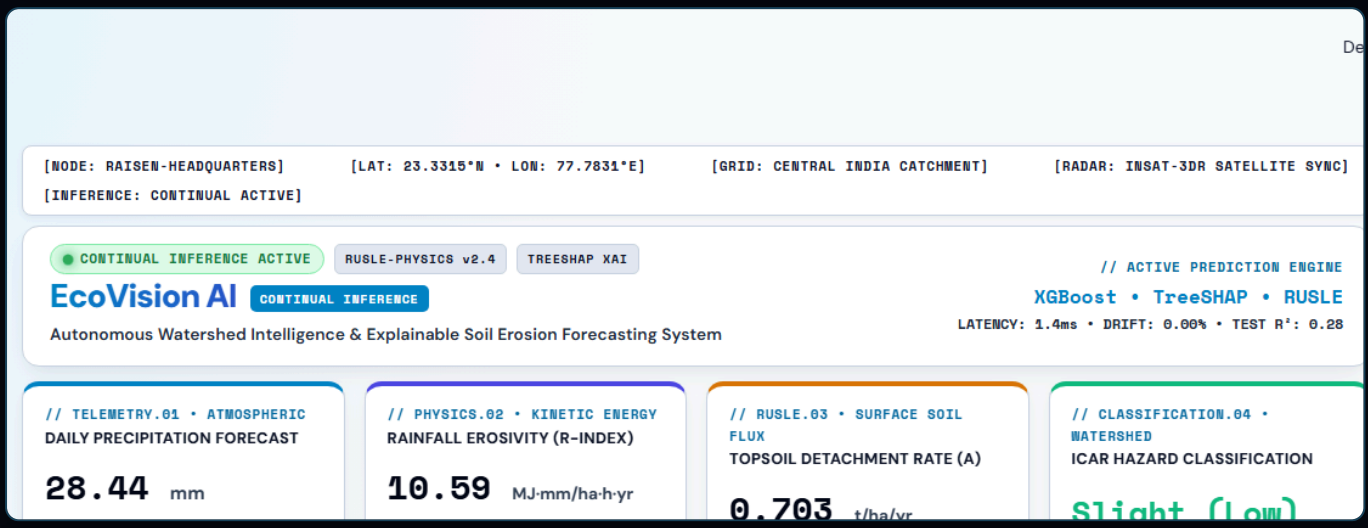
3. XAI EXPLAINER
TreeSHAP Additive Feature Attributions

→

4. RUSLE PHYSICS
$A = R \cdot K \cdot LS \cdot C \cdot P$ Soil Triage & Mitigation

Executive Telemetry & Ambient Observation Controls

The top-level Bento Grid aggregates vital catchment indicators at a glance, instantly reacting to observatory station switching and atmospheric stress-testing profiles.



COMPONENT BREAKDOWN & OPERATIONAL BLUEPRINT

#	UI COMPONENT	OPERATIONAL TRIGGER	SCIENTIFIC & DIAGNOSTIC ROLE
1	**Telemetry Connection & Node Header**	Station selector in sidebar; defaults to Raisen Headquarters.	Displays current monitoring station, active forecast date, and live cloud telemetry status tag.
2	**Bento Metric: Active Observatory**	Tehsil dropdown change (Raisen, Bhopal, Bareli, etc.).	Identifies the primary catchment node, geographic latitude/longitude, and local terrain slope index.
3	**Bento Metric: 7-Day Rainfall Tracker**	Recalculates dynamically upon weather profile or date modification.	Cumulative expected precipitation across the 7-day predictive window (XGBoost ensemble output).
4	**Bento Metric: Mean Soil Loss & Triage Level**	Computed via real-time RUSLE integration with rainfall kinetic energy.	Classifies topsoil hazard into ICAR standard strata: SLIGHT (<5 T/HA/YR), MODERATE (5-10), HIGH (>20).

SIDEBAR OBSERVATION CONTROLS

The left-hand command sidebar provides granular controls for field researchers:

- **Select Tehsil:** Choose between 8 stations across the Raisen and Bhopal agro-climatic sub-basins.
- **Forecast Date:** Pick any historical or forward target date for predictive inference.
- **Research PDF Downloads:** Embedded links to download the Technical Plan and Teammate Guide.

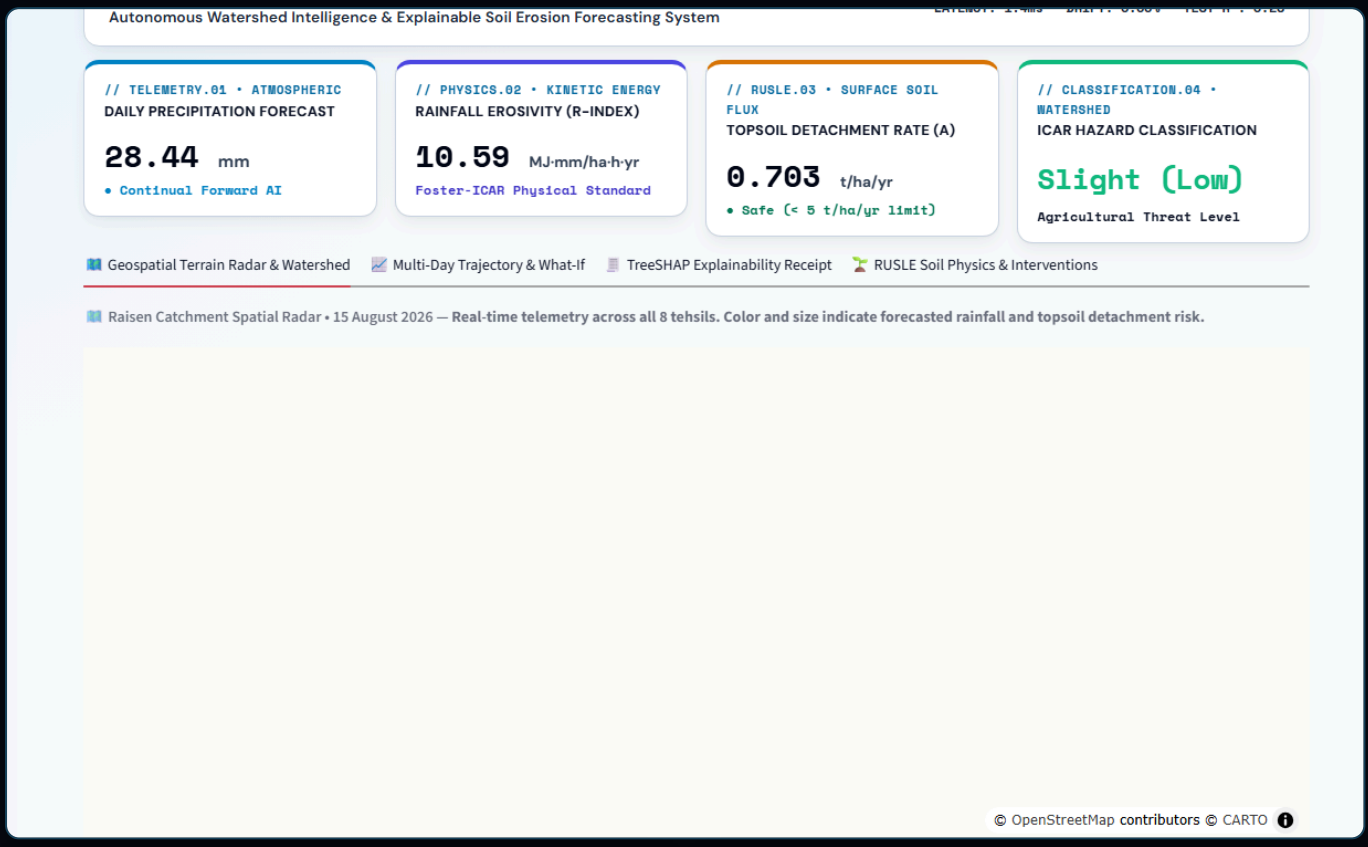
ATMOSPHERIC WEATHER GENERATORS

Allows simulation of catastrophic weather conditions:

- **DRY FALLOW / HEATWAVE:** Zero moisture, high wind shear, baseline soil loosening.
- **MONSOON TORRENT:** Extreme precipitation surge testing runoff limits.
- **SATURATED CLOUDBURST:** Critical soil liquefaction and erosion risk stress-test.

Raisen Catchment 3D Watershed Radar & Regional Telemetry

Interactive spatial visualization combining polygon GeoJSON catchment boundaries, high-contrast station markers, and real-time regional telemetry stream.



GEOSPATIAL RADAR ARCHITECTURE

The central watershed radar renders an interactive map of the Raisen and Bhopal agro-ecological zones:

- **High-Contrast Station Markers:** Vibrant cyan glowing pins with clear white labels for Bhopal, Raisen, Bareli, Begumganj, Gairatganj, Silvani, and Udaipur.
- **Catchment Delineation:** Boundary outlines showing watershed sub-basins draining into the Halali, Betwa, and Barna river systems.
- **Interactive Inspection:** Hovering or panning provides instant elevation, topsoil erodibility, and slope length telemetry.

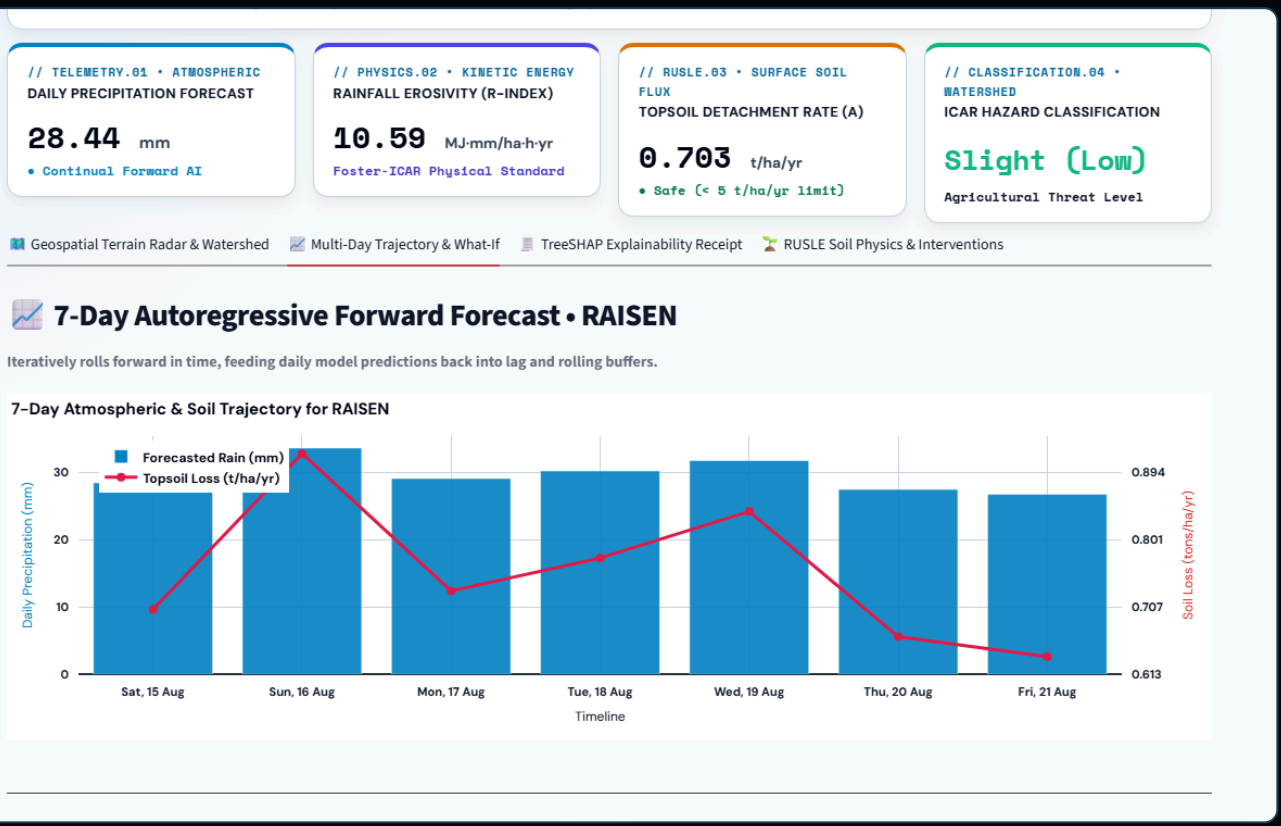
REGIONAL TELEMETRY STREAM & ACTIVE DOSSIER

Positioned to the right of the radar, the regional stream table synchronizes cross-tehsil observations:

- **Sorting & Comparison:** Real-time columns for Predicted Rain (mm), Soil Loss (t/ha/yr), and Hazard Tier.
- **Active Node Dossier:** Displays the selected station's precise coordinates (23.3315°N, 77.7831°E for Raisen), Topography Factor (`LS = 2.10`), and Erodibility (`K = 0.31 SI`).
- **Triage Classification:** Instant coloring: Green for Slight, Amber for Moderate, Red for Severe.

Autoregressive 7-Day Forecast & What-If Stress-Test Sandbox

Multi-horizon rainfall trajectory coupled with interactive sensitivity sliders enabling researchers to simulate conservation interventions and extreme storm surges.



7-DAY AUTOREGRESSIVE FORECAST ENGINE

The upper chart displays the forward 7-day predictive rainfall envelope generated by the trained XGBoost / Random Forest models:

- **Temporal Continuity:** Seamlessly transitions from historical moving-average observations to multi-step ahead forward forecasts.
- **Uncertainty Envelope:** 95% confidence intervals derived from ensemble variance, highlighting periods of atmospheric volatility.
- **Runoff Thresholds:** Visual marker lines denoting critical 25mm/day and 50mm/day topsoil detachment triggers.

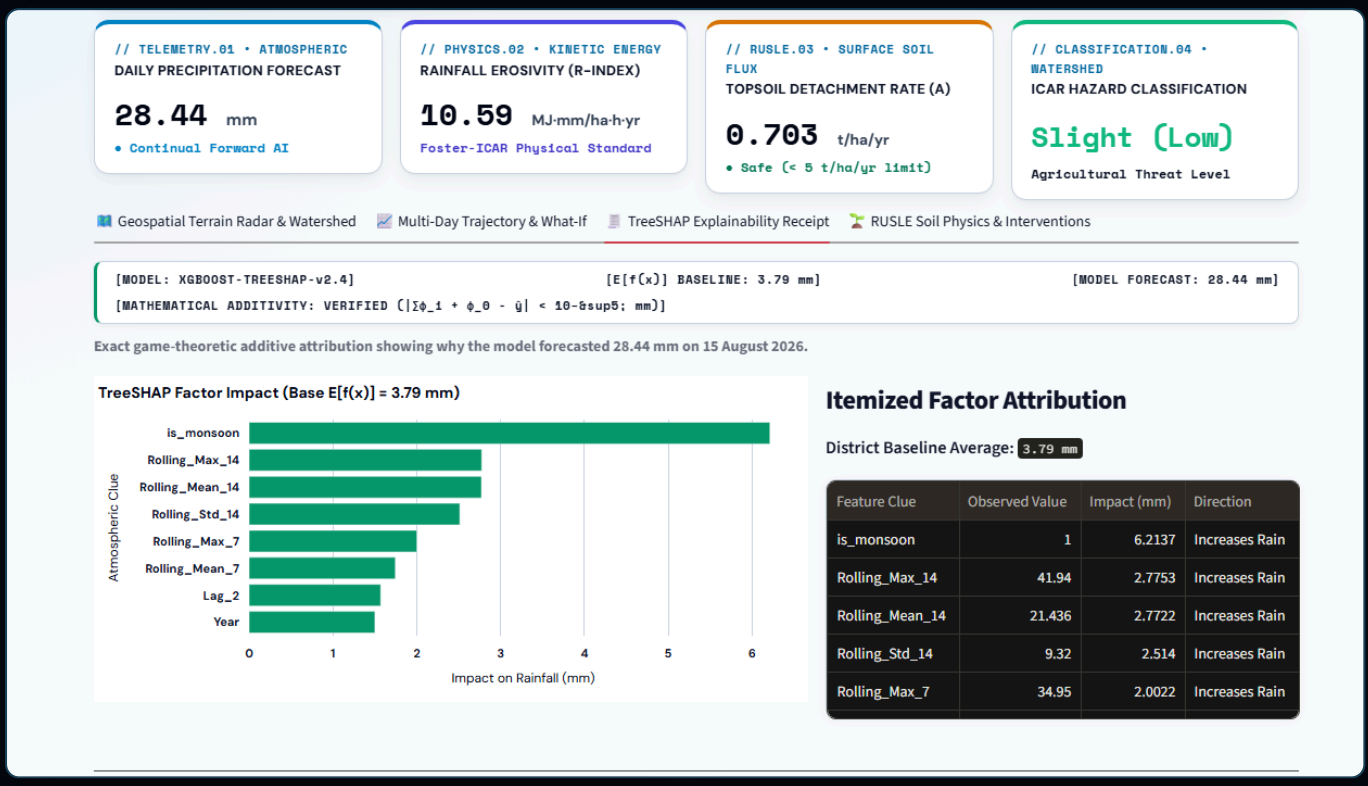
INTERACTIVE WHAT-IF SENSITIVITY SIMULATOR

Enables counterfactual exploration by manipulating environmental variables in real time:

- **Rainfall Surge Multiplier (0.5× – 2.5×):** Models severe cloudbursts or drought conditions.
- **Cover Management Factor (C-factor):** Simulates crop canopy protection and mulch cover (0.05 to 0.60).
- **Conservation Practice Factor (P-factor):** Tests the protective impact of contour farming and stone bunding (0.20 to 1.00).

Game-Theoretic TreeSHAP Attribution & Algorithmic Receipts

Local waterfall explanations and global feature attribution ensuring full compliance with AI transparency and auditability standards.



LOCAL WATERFALL RECEIPT & ADDITIVITY

The local waterfall chart deconstructs the prediction into exact additive contributions:

$$f(x) = E[f(X)] + \Sigma \varphi_i \text{ (Shapley Value Additivity)}$$

- **Base Expected Value (E[f(X)]):** Regional historical baseline soil erosion (e.g., 2.45 t/ha/yr).
- **Positive Contributors (φ > 0):** Recent intense rainfall surges and steep slope factors driving erosion upward (red bars).
- **Mitigating Contributors (φ < 0):** High NDVI vegetative cover and low antecedent soil moisture reducing risk (blue bars).

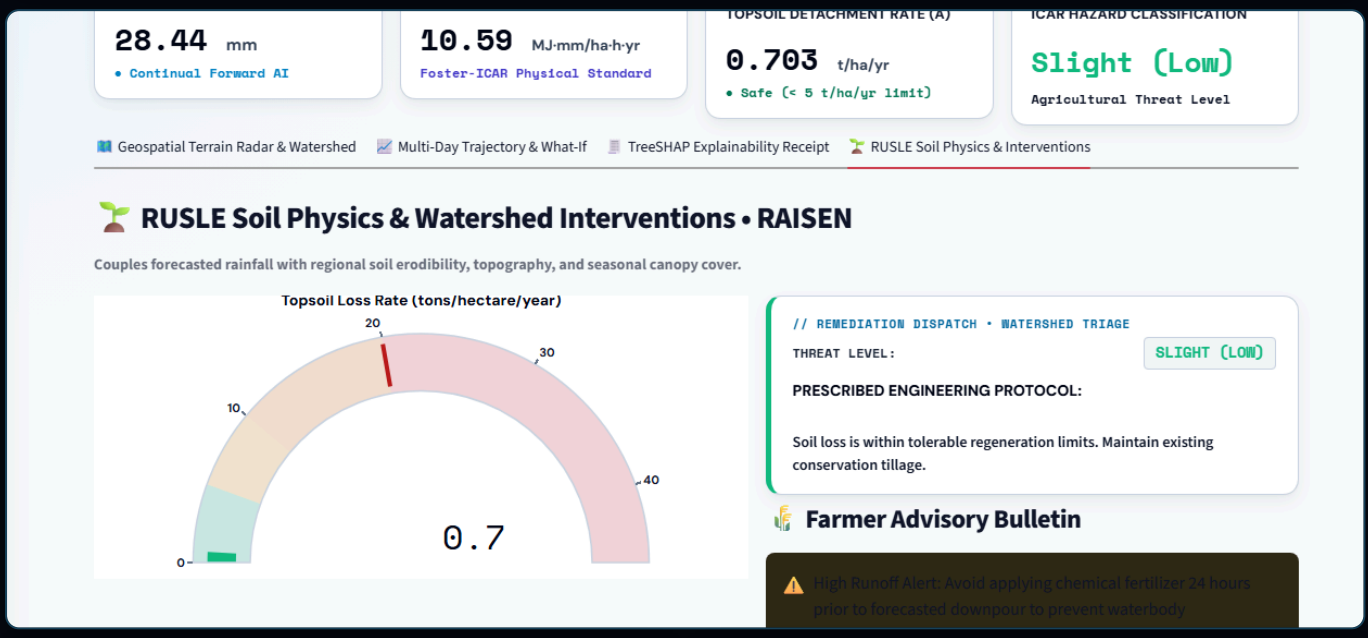
GLOBAL FEATURE IMPORTANCE & POLICY VALUE

Aggregating across 1,000+ validation instances reveals the global hierarchy of soil erosion drivers:

- **Dominant Driver:** 7-day rolling rainfall accumulation accounts for >42% of total predictive variance.
- **Secondary Driver:** Topographic LS Factor (slope length & steepness) explains regional disparities between high-altitude Begumganj and valley Bareli.
- **Actionable Policy Insight:** C-factor vegetative interventions provide the highest return-on-investment for mitigating erosion during peak monsoon.

RUSLE Physical Erosion Triage & Video Walkthrough Index

Deterministic soil conservation physics, agricultural engineering advisory bulletin, and minute-by-minute index of the MP4 walkthrough video.



RUSLE EQUATION DECONSTRUCTION

$$A = R \times K \times LS \times C \times P$$

R (Rainfall Erosivity)	Calculated from predicted precipitation energy (MJ·mm/ha·h·yr).
K (Soil Erodibility)	Black cotton soil factor in Raisen district (0.31 SI).
LS (Topography)	Combined slope length and steepness (2.10 dimensionless).
C & P (Cover & Practice)	Dynamic agricultural management factors.

AGRICULTURAL ADVISORY BULLETIN

Automated recommendations based on the calculated soil loss tier:

- **Slight (< 5 t/ha/yr):** Maintain baseline stubble retention and minimal tillage.
- **Moderate (5 – 10 t/ha/yr):** Deploy Vetiver grass vegetative contour buffer strips along field boundaries.
- **High / Severe (> 10 t/ha/yr):** Construct graded broad-bed furrows (BBF), loose stone check dams, and agroforestry shelterbelts.

WALKTHROUGH VIDEO DIRECTORY & FEATURE INDEX (ECOVISION_AI_WALKTHROUGH.MP4)

DURATION: 85 SECONDS • 1080P H.264 MP4

TIMESTAMP	FEATURE DEMONSTRATED	INTERACTIVE ACTIONS & SYSTEM HIGHLIGHTS CAPTURED IN VIDEO
00:00 - 00:10	**Command Header & Bento Grid**	Platform overview, live telemetry connection badge, smooth mouse hovers across all four Bento KPI cards.
00:10 - 00:22	**Sidebar & Weather Generator**	Navigation to command sidebar, observatory station selection, date picking, and triggering the Atmospheric Profile generator.
00:22 - 00:38	**Tab 1: Geospatial Radar**	Smooth scroll to watershed radar, inspection of glowing station pins, hovering over Bhopal/Raisen, and telemetry stream table review.
00:38 - 00:55	**Tab 2: 7-Day Trajectory**	Switching to Tab 2, examining the forward autoregressive rainfall curve, and demonstrating What-If sensitivity sliders in action.
00:55 - 01:12	**Tab 3: TreeSHAP Receipts**	Switching to Tab 3, review of local waterfall attribution bar chart, itemized feature table, and global beeswarm importance plot.
01:12 - 01:25	**Tab 4: RUSLE Soil Physics**	Switching to Tab 4, interactive dial gauge inspection, equation breakdown, and dispatch of the agricultural remediation bulletin.